

AI CASE STUDY

Fraud Detection Using ML in Digital Banking

Digital Banking



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CLASS NO.

4A2

SUBJECT

Case Studies in AI for Business

Fraud Detection Using ML in Digital Banking

Organization type

A hypothetical mid-to-large Egyptian retail bank offering Meeza and international cards, mobile banking and instant-payment services.

The business promise

Customers expect secure, continuously available and near-real-time payments. The bank must detect suspicious transactions without unnecessary declines or delays.

Primary stakeholders: digital-channel operations, fraud risk, Information Security, customer service, Finance, Compliance, Internal Audit, CBE and customers



Egyptian market pressure and challenges before AI

74.8%

of eligible adults were financially included at year-end 2024

52m

financially included adults at year-end 2024

24/7

instant transfers through Egypt's Instant Payment Network and InstaPay

Challenges before AI

Evolving patterns

Fixed rules struggle to combine stolen credentials, device changes, unusual behavior and rapid fund movement.

Real-time scale

Instant payments shorten the response window, while manual review cannot support every authorization decision.

Fragmented evidence

Card, customer, device, authentication and case data may sit in separate systems, delaying investigation and feedback.

Customer friction

Broad rules can misclassify legitimate changes in channel, location or value and increase complaints.

Business Problem

The Egyptian banks must detect fraudulent card and digital-banking transactions before authorization or fund release, while controlling false declines, unnecessary authentication and investigation workload.

Decision objective

Increase	Confirmed fraud cases and fraudulent value prevented
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Control	False alerts, customer impact and decision latency
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Preserve	Human oversight, reason codes and a complete audit trail
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Why traditional controls were insufficient

TRADITIONAL CONTROL	HOW IT HELPS	WHY THE GAP REMAINS
Fixed limits	Blocks values or velocity above approved thresholds	Inflation and changing behavior can weaken historical thresholds
Blacklist	Blocks known cards, accounts, devices or merchants	Cannot identify a new device, mule account or fraud pattern
OTP / callback	Confirms selected transactions with the customer	Adds delay and may fail against social engineering or compromised access
Manual review	Applies investigator judgment after an alert	Cannot evaluate every digital transaction before completion

Business gaps

Operational

Fragmented evidence and slow response

Strategic

Limited learning from Egyptian fraud outcomes

Financial

Fraud loss, review cost and false declines

Control

Weak decision-to-case traceability

Data types and sources

Data used in the case

Structured

Transaction rows with timestamps, amounts, categories, locations and fraud labels

Historical

Jan 2019 to March 2020 data used for profiling, training and future-period testing

Real time in production

Authorization events and recent customer behavior scored at the point of payment

Unstructured extension

Investigator notes and customer disputes can support NLP analysis after governance review

Production data sources

01 Payment switch or card platform

Authorization event, amount, merchant and channel

02 CRM and customer master

Customer profile, service history and authorized contact

03 Authentication and device logs

Login, MFA, device change and session signals

04 Fraud case management

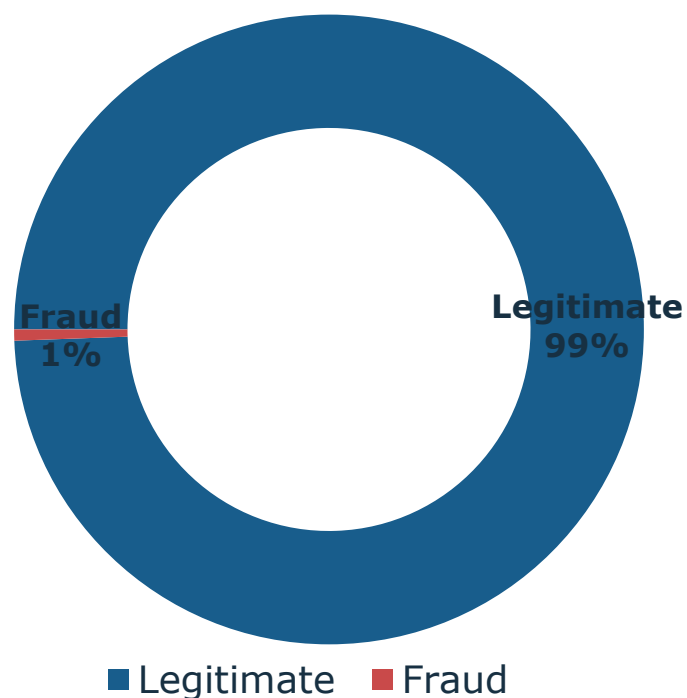
Evidence, investigation outcome and confirmed label

05 ERP Finance

Losses, provisions, recoveries, fees and reconciliation

Dataset profile and analytical readiness

Class distribution



1,048,575 Transactions

215 MB Data Size

6,006 Fraud Cases, equal to 0.573%

23 Source Columns

0 missing values or duplicate rows

Coverage

965 customers 693 merchants 14 categories 51 states

Observation Period January 2019 to March 2020

Analytical implication: use chronological validation, class-sensitive metrics and value capture rather than accuracy alone

AI SOLUTION OVERVIEW

Fraud scoring at the point of payment

A supervised gradient-boosting model estimates fraud risk from transaction, customer, location, category, amount and time features.

The policy layer converts the score into approval, stronger authentication or case review. Confirmed outcomes return as training labels.

Functions affected

Card operations Fraud risk Customer service Finance
Compliance



Machine-learning methods and supporting automation

01 SUPERVISED

Primary method

Learns from confirmed fraud labels. Logistic Regression provides a transparent baseline. Random Forest and gradient boosting capture non-linear interactions.

02 UNSUPERVISED

Complementary signal

Isolation Forest or clustering can surface unusual transactions and emerging patterns. Investigators must confirm whether anomalies represent fraud.

03 SEMI-SUPERVISED

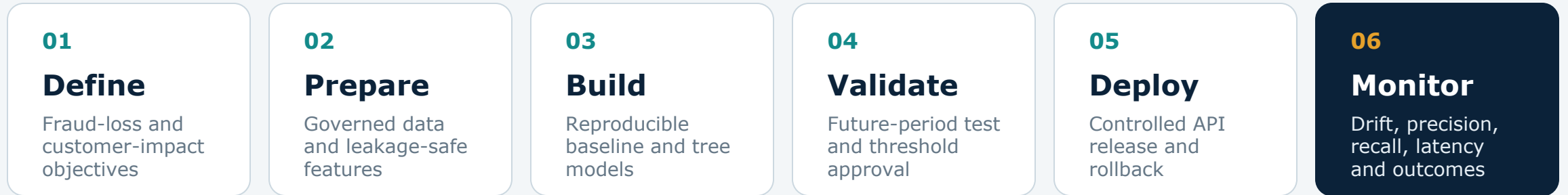
Future extension

Combines a small labelled set with a larger unlabelled population. Use only after label quality and governance controls become mature.

RPA supports the process around the model

Creates fraud cases, collects account and authentication evidence, sends approved customer notifications, and updates recovery or ledger records. Human approval remains mandatory for ambiguous or sensitive decisions.

Secure model development lifecycle



Controls during model development

Chronological 70/15/15 split prevents future information from entering training. Versioned datasets, features, code and approvals create an audit trail.

Controls after deployment

Monthly monitoring compares input distributions and outcomes with approved limits. Material drift, weak alert quality or unexpected customer impact triggers investigation and possible rollback.

Recommended approach: Agile delivery for short iterations and NIST SSDF controls for secure development and release.

Analytical techniques applied to transaction fraud

DESCRIPTIVE

What happened?

Frequency, value and distribution

DIAGNOSTIC

Where is risk concentrated?

Category, hour, amount and distance

PREDICTIVE

Which transaction is likely fraud?

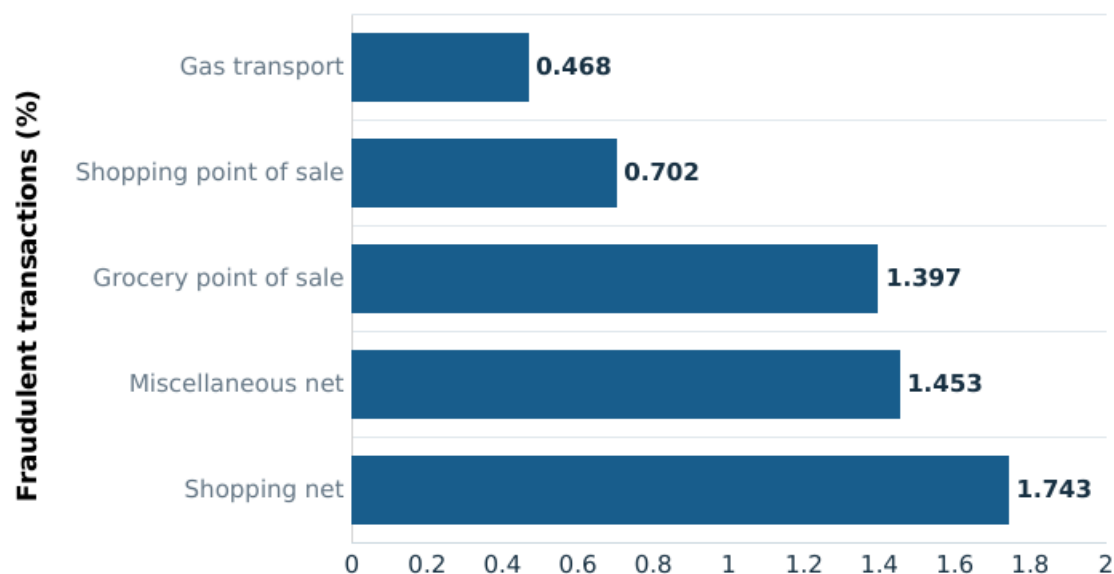
Probability score from trained models

PRESCRIPTIVE

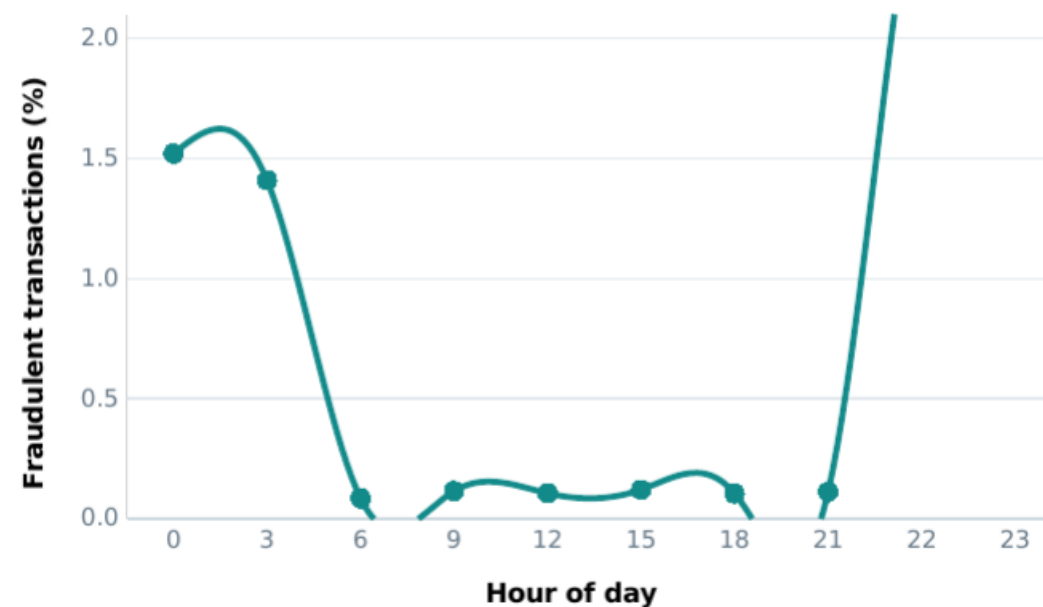
What action fits the risk?

Approve, authenticate or investigate

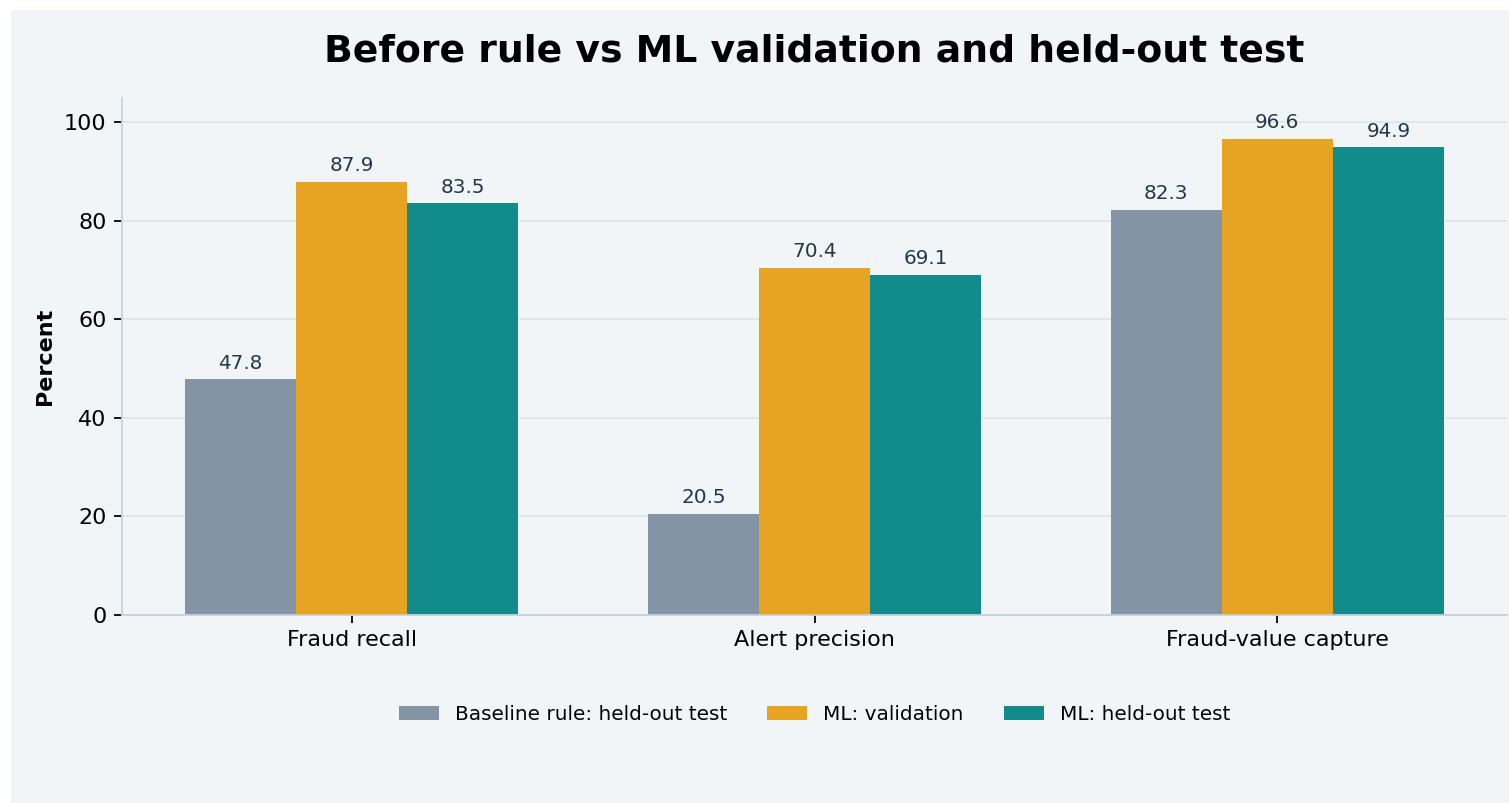
Fraud rate by transaction category



Fraud rate by transaction hour



Model development and held-out test performance



Selected ML model

Histogram Gradient Boosting

83.5%

held-out fraud recall
1,791 of 2,145 cases
detected

69.1%

held-out alert precision
802 false alerts from 2,593
alerts

94.9%

held-out fraud-value capture
\$1,075,351 of \$1,133,325

Development 838,860, validation 209,715, test 555,719, ML
threshold 0.912

Validation: recall 87.9%, precision 70.4%, value capture 96.6%,
PR-AUC 90.7%

Held-out test: PR-AUC 86.7%, ROC-AUC 99.77%, review rate
0.467%

**Decision
design**

Low score: approve

**Medium score: request stronger
authentication**

**High score: investigate or
block under policy**

6. DIGITAL TRANSFORMATION ANALYSIS

AI-enabled fraud operations

Transaction decisions move from periodic rule checks and broad alert queues to continuous risk scoring at the point of payment.

Operating model

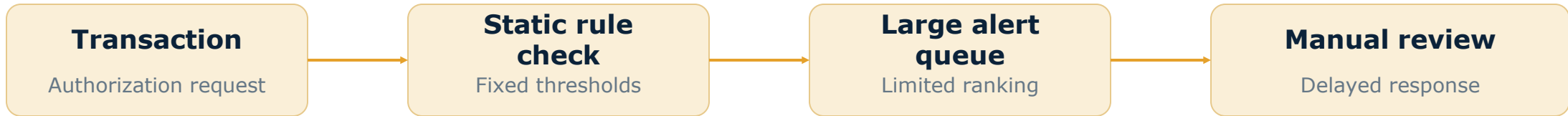
The model ranks risk. Policy selects a proportionate response. Investigators retain authority over ambiguous and high-impact cases. Confirmed outcomes improve future detection.

Transformation goal: faster protection with less unnecessary customer friction

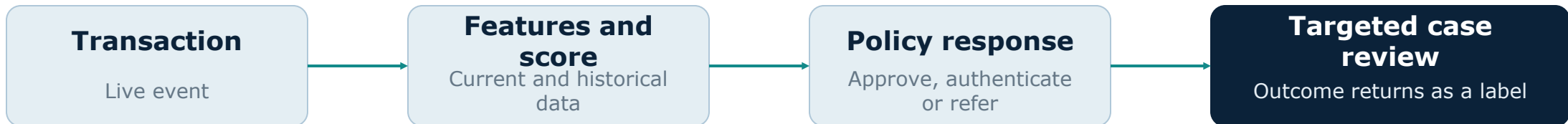


Fraud operations before and after AI

BEFORE: rule-heavy and reactive



AFTER: risk-based and near real time



Process

Continuous scoring replaces broad batch triage

Automation

RPA creates cases, gathers evidence and updates approved records

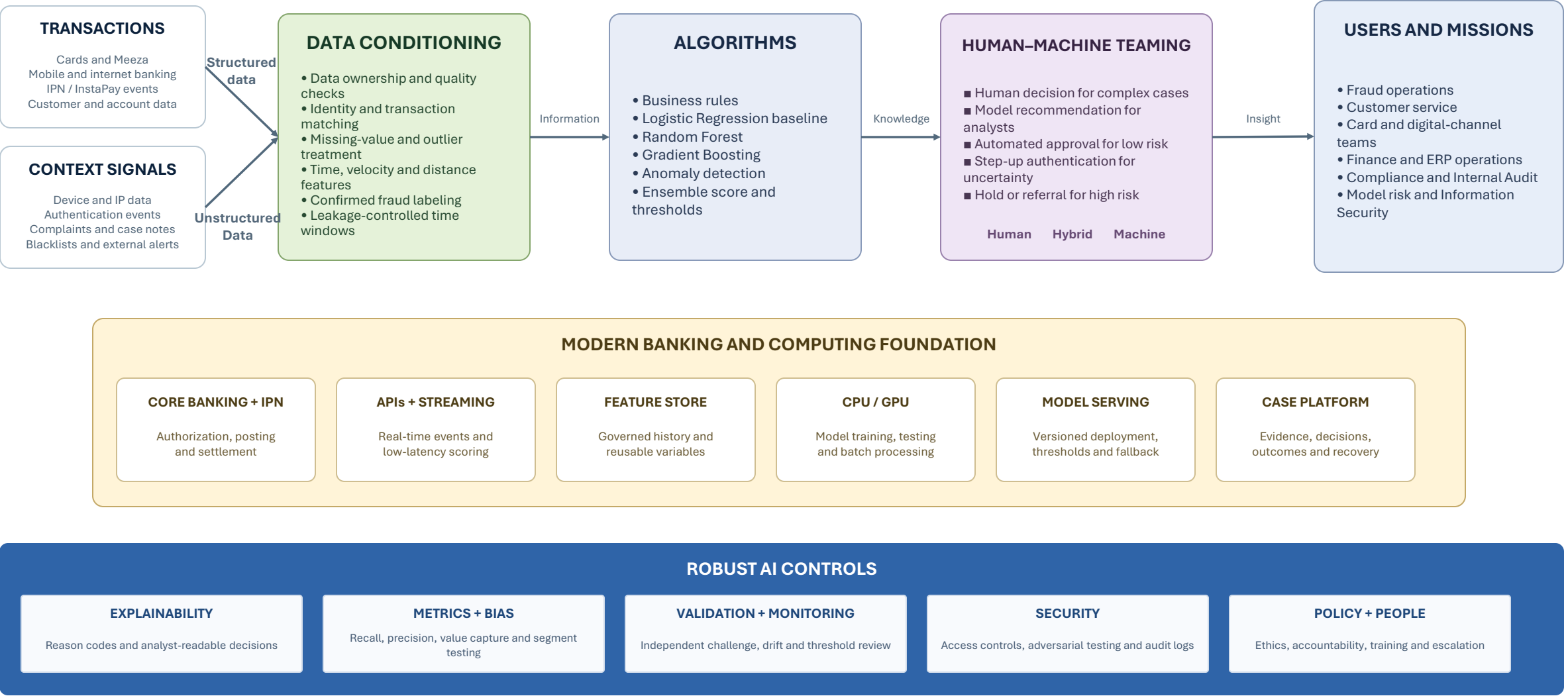
Culture

Analysts focus on complex investigations and model feedback

Customer experience

Low-risk payments pass quickly while medium risk receives step-up authentication

ERP-AI architecture and data flow



Modules, benefits and implementation challenges

MODULE	ROLE IN THE SOLUTION	KEY CONTROL
Payment switch	Authorization event and response	Latency, availability, fail-safe rules and decision timestamps
AI and feature platform	Scoring, versioning and monitoring	Approved model, threshold, features, lineage and drift monitoring
ERP Finance / GL	Losses, provisions, fees and recovery	Transaction-to-ledger reconciliation
CRM	Customer contact and service recovery	Authorized scripts and consent records
Case management / SIEM	Evidence, outcomes, reviews, overrides and security monitoring	Segregation of duties, logs, escalation and override reasons

Integration benefits

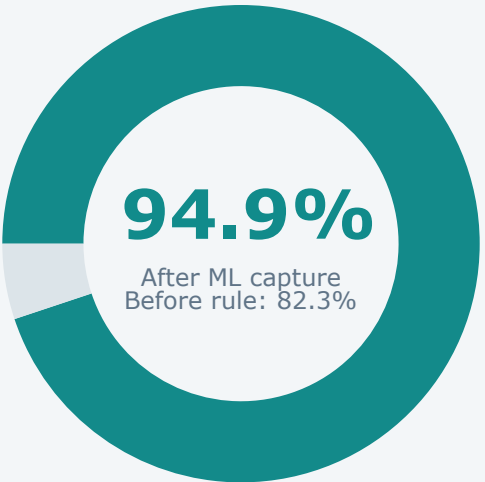
One case reference across systems, faster recognition and recovery, automated reconciliation, and measurable decision latency and reviewer outcomes

Implementation challenges

Inconsistent identifiers, delayed fraud labels, missing decision and override data, API latency, batch ERP interfaces, customization cost and unclear ownership

Measured proof-of-concept value and the ROI framework

After-implementation
fraud-value capture



■ Detected value ■ Missed value

5,004 to
2,593

flagged transactions: before to after
from 555,719 held-out test transactions

1,025 to
1,791

true fraud alerts: before
to after

3,979
to 802

false alerts:
before to
after

0.900% to
0.467%

review rate: before to after

ROI outcome

Net benefit: \$129,992 ROI: 130.0%
Assumes \$5 per review, \$25k annual run and \$100k
implementation

KPIs

Recall 47.8% to 83.5% Value capture 82.3% to 94.9% Precision 20.5% to 69.1% FPR 0.719% to 0.145%

ROI is illustrative: decision latency and overrides are unmeasured; replace assumed review, running and implementation costs with bank actuals

Before = amount rule flagging transactions above the training 99th percentile (\$544.89). After = Histogram Gradient Boosting at validation-selected threshold 0.912334. ROI uses \$5 per review, \$25,000 annual model running cost and \$100,000 implementation cost. Under a first-year cash view that also deducts implementation cost, the remaining surplus is approximately \$29,992.

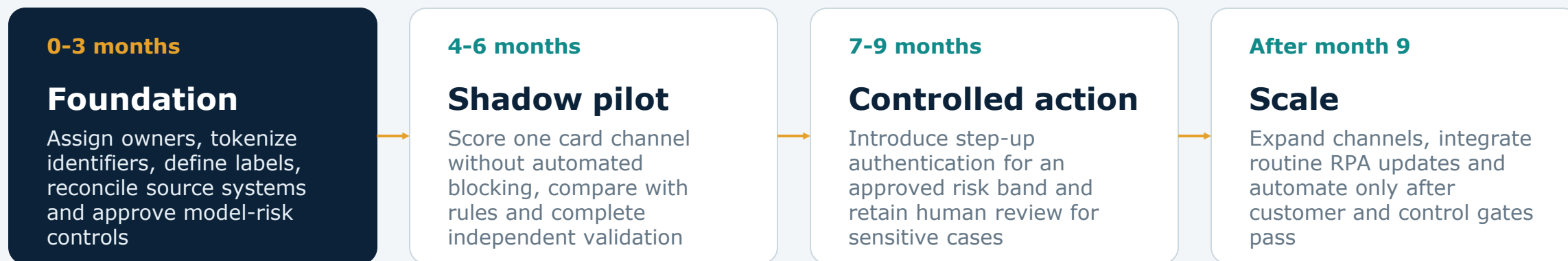
Risk, regulation and information-security controls

RISK	CONTROL RESPONSE
Privacy and excessive data use	Lawful basis, purpose limitation, tokenization, minimization, access controls and retention limits
Bias and unfair customer friction	Compare error rates across defensible groups, challenge proxy features and provide appeal or human review
Model drift	Monitor population, scores and outcomes with approved thresholds and retraining triggers
Cyberattack or manipulation	Secure APIs, authentication, SIEM monitoring, integrity checks and incident response
Automation error and weak explainability	Human-in-the-loop thresholds, reason codes, reconciliation and complete audit logs

Selected ISO/IEC 27001:2022 control Annex A 8.16 - Monitoring activities

Evidence should include approved scenarios, time-synchronized logs, alert ownership, escalation records, periodic tuning and proof that monitoring remains protected.

Phased implementation roadmap



Future analytical scope

Richer features

Customer velocity, device intelligence, authentication behavior and chargeback timing

Graph analytics

Link cards, devices, addresses, merchants and beneficiaries to identify coordinated rings

Complementary learning

Test anomaly detection and semi-supervised learning after label governance matures

Decision optimization

Use amount-sensitive thresholds, expected-loss costs and champion-challenger monitoring

Governance gate: no autonomous blocking until real-data validation demonstrates stable performance, fairness and recoverable customer outcomes